Activity 7: Reliability Failure Analysis

Course: Semiconductor Device and Process Integration Masterclass

Course code: C605

Duration: 45 minutes

PPT alignment: Trainer Slides v1.0, concept slides 356-366 and Activity 7 overview slide 422.

Workplace scenario

Qualification finds early-life gate failures and resistance drift in upper-metal interconnects after accelerated stress.

Goal

Verify whether TDDB, hot-carrier degradation or electromigration best explains the evidence.

Files in this folder

- README.md / README.pdf - complete procedure and acceptance criteria.
- SCENARIO.md / SCENARIO.pdf - facts, assumptions and role brief.
- WORKSHEET.md / WORKSHEET.pdf - structured learner response.
- EVIDENCE-CHECKLIST.md / EVIDENCE-CHECKLIST.pdf - completion and review record.
- Any CSV file in this folder - supplied teaching data for analysis.

Step-by-step procedure

1. Classify each symptom by structure, stress condition, time dependence and electrical signature.
2. Compare the evidence against TDDB, hot-carrier and electromigration mechanism profiles.
3. Select the confirmatory stress test and physical-analysis method for each leading hypothesis.
4. Check whether the observed distribution supports a random defect, systematic process shift or wear-out mechanism.
5. Write a verification statement that names the supported mechanism and remaining evidence gap.

Required evidence

Failure-mechanism comparison and reliability verification statement.

Include your completed worksheet, any calculations or annotated diagrams, the evidence checklist, and a short note identifying assumptions or unresolved evidence gaps.

Acceptance criteria

[ ] All three reliability mechanisms are differentiated

[ ] Selected tests match the suspected structure

[ ] The conclusion identifies an evidence gap

Verification

Before submission, ask a peer to challenge one causal link. Revise the conclusion if the challenge reveals that evidence and inference have been mixed. Your final response must distinguish: observed fact, interpretation, decision, and follow-up check.

Troubleshooting

- If several mechanisms fit the symptom, choose a measurement that discriminates between them.
- If a process module lacks a defined input, output or control gate, the flow is incomplete.
- If an action is not tied to a verified cause, record it as a hypothesis test rather than corrective action.
- If calculations disagree, show the method and units so the discrepancy can be reviewed.